

MANAV GOEL

me.manavgoel@gmail.com | [+91 83446 03000](tel:+918344603000) | github.com/Manav0411 | linkedin.com/in/manavgoel1104

Education

University School of Information, Communication and Technology, GGSIPU

2023 – Present

B.Tech in Computer Science Engineering | CGPA: 8.1/10

New Delhi, India

Experience

AI Engineer Intern

Jun 2026 – Aug 2026

IntelGrader

Remote

- Built a Python document-processing pipeline for AI-based assessment: ingested and chunked 50+ educational documents into a ChromaDB vector store, then served retrieval through hybrid semantic search, BM25 reranking and LLM-based query routing to reach 95%+ chapter-classification accuracy.
- Automated batch answer generation and evaluation for 1200+ exam questions using structured retrieval, evaluation and grading workflows, reaching up to 92% benchmark accuracy and cutting manual content-review effort.
- Engineered a multi-agent content pipeline (Python LLM orchestration, Node.js, browser automation) converting Class 8–12 STEM content into interactive webpages across 12 visual families, with automated validation and self-repair that retries failed generations and cut recurring failures across 25+ outputs.

Backend Developer Intern

Dec 2025 – Feb 2026

Hyrte

New Delhi, India

- Restructured the LangChain prompt orchestration layer behind the interview service's FastAPI endpoints, improving conversational coherence and removing repetitive system responses across multi-turn sessions.
- Reduced LLM token usage by 42% and improved API response latency by restructuring prompts and adding rolling context windows in the FastAPI backend; validated endpoints with Postman.

Projects

InsightGraph – Async Briefing Service | Python, FastAPI, PostgreSQL, LangGraph, Next.js | [GitHub](#) | [Live](#)

- Built the FastAPI backend for a daily AI briefing product: 15 REST endpoints across 7 routers over a 5-table PostgreSQL schema (SQLAlchemy, Neon serverless), with Clerk JWT auth verified against remote JWKS, a webhook that syncs user records on signup, per-user data isolation, and pgvector columns for semantic lookup of past briefings.
- Made a 10+ minute LLM pipeline safe to run behind an HTTP API: an async endpoint returns a task ID immediately via FastAPI background tasks, the client polls a status endpoint for stage-level progress, and task state is persisted so jobs left running after a restart are reconciled to failed instead of hanging the UI.
- Cut run time and cost with bounded concurrency (semaphore-limited `asyncio.gather`), fail-open handling of HTTP 429 that removed a 6-minute rate-limit stall, and tiered model routing (small model for filtering, large for analysis); scheduled daily runs with a GitHub Actions cron, pinned every dependency, and gated merges on a mocked pytest suite spanning units, services, routes and the full pipeline.

Groundwork – Cited Q&A Service | Python, FastAPI, PostgreSQL, Docker, AWS EC2, LangGraph, Next.js | [GitHub](#) | [Live](#)

- Built the FastAPI backend for a cited engineering Q&A service: 20 REST endpoints over an 8-table PostgreSQL schema (async SQLAlchemy, Alembic) with GIN full-text and pgvector HNSW indexes. Exact questions route to typed SQL and answer in under 25 ms with no model call; open-ended ones go to hybrid search fused by reciprocal rank fusion, tuned against a labeled eval set.
- Designed the data ingestion layer for GitHub, Jira and Slack around a per-source sync state machine: cursor-based incremental syncs, a lock preventing concurrent runs, failed syncs that never advance the cursor, API pagination, and content-hash upserts that re-embed only changed documents; added an HMAC-verified webhook after polling silently skipped backdated commits.
- Deployed and hardened the service in production: Docker on AWS EC2 behind Caddy/TLS, Next.js on Vercel, API key auth, sliding-window rate limiting, JSON logs with correlation IDs and a Prometheus metrics endpoint; cut response time from ~68 s to 1.8 s and gated releases on ~480 unit and Postgres integration tests plus a CI schema-drift check.

Technical Skills

- Languages:** Python, SQL, JavaScript, TypeScript, C++
- Backend & Databases:** FastAPI, Django, REST APIs, Node.js, SQLAlchemy, Alembic, Pydantic, JWT/OAuth, PostgreSQL, pgvector
- Frontend:** React, Next.js, Tailwind CSS, HTML, CSS
- Tools & DevOps:** Docker, AWS EC2, GitHub Actions (CI/CD), Git, Caddy, Vercel, pytest, Postman, Prometheus
- AI/ML:** LangGraph, LangChain, RAG, hybrid retrieval, reranking (RRF), vector search, embeddings, structured outputs, Groq, Gemini, HuggingFace, ChromaDB, LangSmith, DeepEval
- CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements

- Solved 700+ DSA problems across LeetCode, GeeksforGeeks and CodeStudio ([Profile](#)); Level 3 Conqueror, Coding Ninja Slayground 2.0 ([Certificate](#)).

Positions of Responsibility

Lead, ACM ICPC Club | USICT, GGSIPU

2024 – 2026

- Organized coding contests and DSA workshops, driving competitive programming engagement across campus.
- Mentored 150+ junior students in algorithms and problem-solving, boosting participation in national coding competitions.